# Explainability for CNN-Based Image Classification: A Comparison of Grad-CAM and Integrated Gradients

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Explainable AI – Final Assignment

**General topic:** Model-specific explainability
**Use case/domain:** CNN-based image classification
**Dataset:** Fashion-MNIST (TensorFlow Datasets)
**Dataset link:** `https://www.tensorflow.org/datasets/catalog/fashion_mnist`
**Methods:** Grad-CAM and Integrated Gradients
**Research question:** To what extent do Grad-CAM and Integrated Gradients highlight meaningful image regions for correct and incorrect CNN predictions on Fashion-MNIST?

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## 1 Introduction

Convolutional Neural Networks (CNNs) are widely used for image classification because they can learn hierarchical visual patterns directly from pixel inputs. Despite their strong predictive performance, CNNs are often criticized as "black boxes" because it is difficult to understand why a specific prediction was made. This lack of transparency can be problematic when users want to inspect model behavior, understand errors, or evaluate whether predictions rely on meaningful visual evidence rather than irrelevant correlations [1].

Explainable Artificial Intelligence (XAI) addresses this issue by providing methods that make model decisions more understandable. In image classification, explanation methods typically highlight image regions or input features that contributed most to the model output. These explanations can help determine whether a model focuses on relevant object characteristics, relies on spurious cues, or fails in understandable ways [1, 2].

This report examines two different model's explainability for CNN-based image classification using the Fashion-MNIST dataset. Fashion-MNIST is a benchmark dataset of grayscale clothing images. Although the dataset is not used for significant real-life decisions, it is a useful controlled setting for comparing explanation methods because the images are visually simple, class labels are clear, and predictions are relatively easy to inspect [3].

The two methods compared in this report are Grad-CAM and Integrated Gradients. While both aim to explain CNN predictions, they do so differently: Grad-CAM highlights important spatial regions using convolutional feature maps, whereas Integrated Gradients assigns attribution values directly to the input. Because of this difference, the methods may vary in clarity, level of detail, and usefulness for interpreting correct and incorrect predictions.

The research question of this report is therefore: *To what extent do Grad-CAM and Integrated Gradients highlight meaningful image regions for correct and incorrect CNN predictions on Fashion-MNIST?*

In this context, *meaningful image regions* refers to regions that plausibly correspond to the clothing item being classified and help explain why the model predicted a certain class. For example, an explanation for an ankle boot is more meaningful if it highlights the sole or upper boot shape than if it highlights mostly blank background pixels. This question is especially relevant because explanation methods do not always agree, and visually convincing explanations are not necessarily the most informative or faithful ones.

# 2 Methods

## 2.1 Use Case and Dataset

The selected use case is CNN-based image classification on Fashion-MNIST. The dataset contains grayscale images of ten clothing categories: T-shirt/top, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag, and ankle boot. It consists of 60,000 training images and 10,000 test images, each with a resolution of $28 \times 28$ pixels [3].

Fashion-MNIST is suitable for this project because it is computationally lightweight, easy to implement, and visually interpretable. Since each image contains one centrally placed object on a plain background, it is easier to inspect whether explanation methods highlight relevant parts of the item.

## 2.2 CNN Model

A CNN was used as the classification model. The network processes Fashion-MNIST images through convolutional layers that learn local visual patterns such as edges and contours. Pooling layers reduce spatial dimensionality, and fully connected layers transform the learned representation into class probabilities.

Before training, pixel values were normalized to the range $[0, 1]$, and the images were reshaped to include a channel dimension. The model was trained for five epochs with the Adam optimizer and a batch size of 64. The model was then evaluated on the test split.

The main goal of the CNN in this project was not to achieve the best possible performance, but to serve as a working model for applying and comparing the explanation methods.

### 2.3 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) explains predictions by identifying which spatial regions in an image were most influential for the predicted class. It computes gradients of the target class score with respect to the feature maps of a selected convolutional layer, usually the last one. These gradients are averaged to obtain a weight for each feature map channel, and the weighted feature maps are combined into a heatmap [1].

Grad-CAM is well suited to CNNs because it preserves spatial information and usually produces intuitive visual explanations. Its main advantage is that it highlights broad regions of interest in a form that is easy to inspect. However, its output is less detailed and can change depending on the layer used.

### 2.4 Integrated Gradients

Integrated Gradients is an attribution method that assigns an importance value to each input feature. It estimates feature importance by integrating gradients along a path from a baseline input, often a black image, to the actual input image. In image classification, this produces pixel-level attributions showing which parts of the input contributed most to the prediction [2].

A key strength of Integrated Gradients is that it gives detailed attributions because it works directly on the input image. This can make it more precise than Grad-CAM. However, the visualizations can also be noisier and may change depending on the chosen baseline and the number of steps used.

### 2.5 Evaluation Approach

The explanation methods were evaluated mainly by visually comparing them. For selected test images, the original image, predicted label, and the Grad-CAM and Integrated Gradients visualizations were shown side by side. The analysis included both correct and incorrect predictions.

The comparison was guided by four criteria. First, *interpretability*: whether the explanation is visually understandable. Second, *localization*: whether the highlighted regions correspond to relevant parts of the clothing item. Third, *error analysis value*: whether the explanation helps reveal why a misclassification occurred. Fourth, *consistency*: whether visually similar inputs yield similar explanation patterns.

## 3 Results

### 3.1 Classification Results

The CNN achieved a test accuracy of 0.8944 on Fashion-MNIST, with a test loss of 0.2950. This indicates that the model learned the dataset well enough to provide a meaningful basis for

applying and comparing explanation methods.

Table 1: Classification performance of the CNN on Fashion-MNIST.

| Metric | Value |
|---|---|
| Test accuracy | 0.8944 |
| Test loss | 0.2950 |
| Epochs | 5 |
| Batch size | 64 |

### 3.2 Explanation Examples

Figure 1 presents a correctly classified ankle boot example, while Figure 2 presents an incorrectly classified ankle boot that was predicted as a sandal. For each case, the original image is shown together with Grad-CAM and Integrated Gradients.

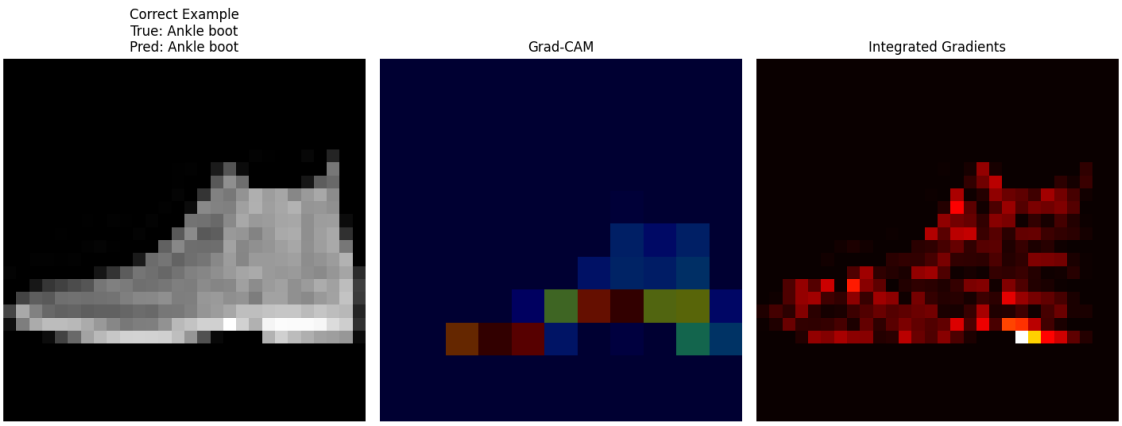


Figure 1: Correctly classified ankle boot with Grad-CAM and Integrated Gradients explanations.

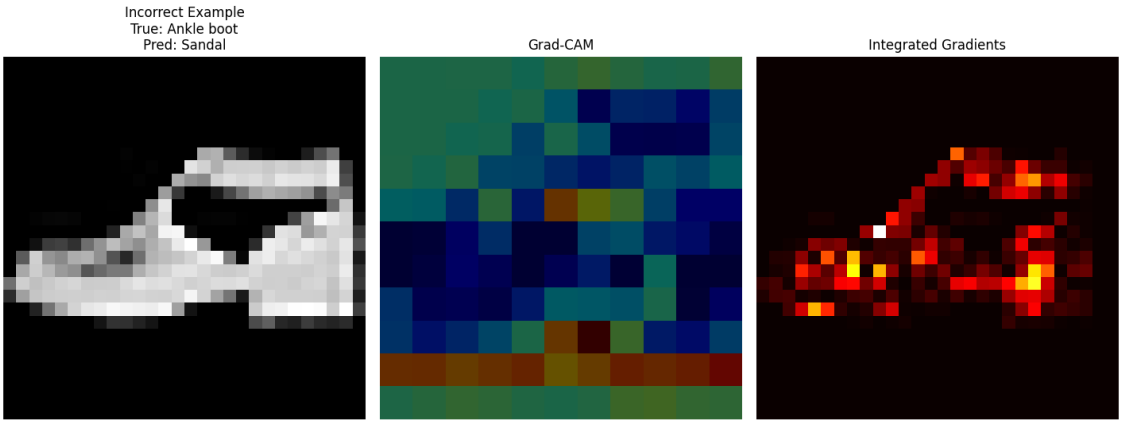


Figure 2: Incorrectly classified ankle boot predicted as sandal, with Grad-CAM and Integrated Gradients explanations.

Overall, Grad-CAM produced broader and less detailed heatmaps, while Integrated Gradients gave more detailed attribution patterns. In the correctly classified example, both methods highlighted relevant parts of the ankle boot instead of the background. However, Grad-CAM focused on the object more generally, while Integrated Gradients focused more on details such as the contours and the sole.

In the incorrect example, where an ankle boot was misclassified as a sandal, the difference between the methods became even clearer. Integrated Gradients still followed the shape of the object in a relatively detailed way, especially around the sole and upper contour. Grad-CAM, by contrast, appeared less focused and spread attention more broadly across the image. This suggests that Grad-CAM may be more useful for general inspection, whereas Integrated Gradients may provide more insight for fine-grained error analysis.

## 4 Discussion

The comparison shows that Grad-CAM and Integrated Gradients explain CNN predictions in different ways. Grad-CAM is easier to understand when the goal is to see the general area the model focused on. In Fashion-MNIST, this is useful for checking whether the model looks at the clothing or the background. However, Grad-CAM can be too broad when small differences between similar classes such as shirts, coats, and pullovers are important.

Integrated Gradients, on the other hand, provides more detailed attribution patterns, such as contours, edges, and localized shape information. This makes it especially useful for under-

standing small differences between classes and for analyzing misclassifications. In the examples in this report, it was more informative for the incorrect prediction because it still highlighted important object features even when the model confused an ankle boot with a sandal. However, its visualizations were also noisier and harder to interpret quickly than Grad-CAM".

Several pitfalls should be considered. First, explanations that looked convincing do not always reflect how the model actually made its decision. Second, different methods can give different explanations for the same prediction, which can make them harder to interpret. Third, users may trust heatmaps too much and think they are true explanations, even though they just show patterns learned by our model and finally explanation methods don't automatically make a model robust or reliable.

A limitation of this study is that Fashion-MNIST is a relatively simple benchmark rather than a complex real-world problem. This makes the comparison easier to control and visualize, but it also means the results cannot be fully generalized. Still, the dataset is useful for comparing how explanation methods differ in clarity, localization, and usefulness for understanding predictions. Overall, the results suggest that Grad-CAM and Integrated Gradients should be seen as complementary methods rather than competing ones: one is better for broad visual localization, while the other is better for detailed attribution.

# 5 Conclusion

This report compared Grad-CAM and Integrated Gradients for explaining CNN-based image classification on Fashion-MNIST. The goal was to examine how well these methods highlight meaningful image regions for correct and incorrect predictions. The results showed that both methods generally focused on relevant parts of the clothing item, but they differed in detail and usefulness

Grad-CAM produced broader and more intuitive heatmaps, making it useful for general visual inspection of where the network was attending. Integrated Gradients provided more fine-grained attribution patterns and was especially informative in the misclassified example, where it highlighted object-specific features more clearly than Grad-CAM. These findings suggest that the two methods offer complementary perspectives on CNN behavior rather than serving exactly the same purpose.

Even in a simple benchmark setting, the comparison illustrates that explanation quality depends not only on the method itself, but also on the type of task and the goal of interpretation. For practical use, Grad-CAM may be preferable for quick overview, while Integrated Gradients may be more useful for detailed error analysis.

## Code Availability

The code used for training the CNN and generating Grad-CAM and Integrated Gradients explanations is available at: `https://github.com/s1100458/fashion-mist-xai`.

## References


[1] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2020). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *International Journal of Computer Vision, 128*(2), 336–359.

[2] Sundararajan, M., Taly, A., & Yan, Q. (2017). Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning* (pp. 3319–3328).

[3] Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*.